Codebook for: Democratization, Elections, and Public Goods: The Evidence from Deforestation

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This codebook lists each variable in the analysis dataset and traces it to the dataset from which it originated. The datasets used to construct the analysis datasets are:

Global Administrative Boundaries, accessed March 2018
https://gadm.org/

GADM maps and data. Accessed March 4, 2018. https://gadm.org/data.html

Global 0.05 decimal degree forest cover maps from 1982-2016
https://www.nature.com/articles/s41586-018-0411-9

Hansen, M., X. Song. Vegetation Continuous Fields (VCF) Yearly Global 0.05 Deg. 2018, distributed by NASA EOSDIS Land Processes DAAC, https://doi.org/10.5067/MEaSUREs/VCF/VCF5KYR.001. Accessed 2020-12-27.

Global Ecological Zones
http://www.fao.org/geonetwork/srv/en/metadata.show?id=1255

Team, GeoNetwork. 2007. "GeoNetwork Opensource Portal to Spatial Data and Information." InteractiveResource. Global. February 7, 2007. http://www.fao.org/geonetwork/srv/en/metadata.show?id=1255.

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https://www.systemicpeace.org/inscrdata.html

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Database of Political Institutions, accessed March 2018
https://datacatalog.worldbank.org/dataset/wps2283-database-political-institutions

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Varieties of Democracy, accessed March 2020
https://www.v-dem.net/en/

Coppedge, Michael, John Gerring, Carl Henrik Knutsen, Staffan I. Lindberg, Jan Teorell, David Altman, Michael Bernhard, M. Steven Fish, Adam Glynn, Allen Hicken, Anna Lu¨hrmann, Kyle L. Marquardt, Kelly McMann, Pamela Paxton, Daniel Pemstein, Brigitte Seim, Rachel Sig- man, Svend-Erik Skaaning, Jeffrey Staton, Steven Wilson, Agnes Cornell, Nazifa Alizada, Lisa Gastaldi, Haakon Gjerløw, Garry Hindle, Nina Ilchenko, Laura Maxwell, Valeriya Mechkova, Juraj Medzihorsky, Johannes von R¨omer, Aksel Sundstr¨om, Eitan Tzelgov, Yi-ting Wang, Tore Wig, and Daniel Ziblatt. 2020. ”V-Dem [Country–Year/Country–Date] Dataset v10” Varieties of Democracy (V-Dem) Project. https://doi.org/10.23696/vdemds20.
and:
Pemstein, Daniel, Kyle L. Marquardt, Eitan Tzelgov, Yi-ting Wang, Juraj Medzihorsky, Joshua Krusell, Farhad Miri, and Johannes von R¨omer. 2020. “The V-Dem Measurement Model: La- tent Variable Analysis for Cross-National and Cross-Temporal Expert-Coded Data”. V-Dem Working Paper No. 21. 5th edition. University of Gothenburg: Varieties of Democracy Institute.

World Development Indicators, accessed March 2018
Codes: "NV.AGR.TOTL.ZS", "SL.AGR.EMPL.ZS", "SP.POP.GROW", "SP.POP.TOTL", "NY.GDP.PCAP.KD.ZG", "NY.GDP.PCAP.KD"
https://databank.worldbank.org/reports.aspx?source=world-development-indicators

World Development Indicators. Washington, D.C. March 4, 2018. :The World Bank. https://datacatalog.worldbank.org/dataset/world-development-indicators

International Political Economy Data Resource
https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/28003

Graham, Benjamin A.T.; Tucker, Jacob R., 2016, "The International Political Economy Data Resource", https://doi.org/10.7910/DVN/X093TV, Harvard Dataverse, V4

Cell Level Characteristics

x

Longitude of the center of the pixel

y

Latitude of the center of the pixel

area

Area of the pixel, varies based on its latitude with pixels near the equator having an area of 30.25 square kilometers, decreasing as pixels approach 90 or -90 degrees

forest

Percent of each cell covered by forest, defined as vegetation 5 meters or taller. Extracted from Hansen and Song 2018.

forest.l

Percent of each cell covered by forest in the previous year. Extracted from Hansen and Song 2018. NA if data are not available.

forest.diff

Change in percent forest cover from the previous year to the current year. Extracted from Hansen and Song 2018. NA if data are not available.

nn_forest

Average forest cover of rook's case neighboring cells (cells sharing an edge). Generated using Hansen and Song 2018. NA if data are not available.

nn_forest.l

Average forest cover of rook's case neighboring cells in the previous year. Generated using Hansen and Song 2018. NA if data are not available.

FID

Unique cell identifier, generated by the author

Location Characteristics

year
: Year of the observation

Countryeng
: Name of the country, from GADM

GID_1
: Level 1 administrative unit code, from GADM

GID_2
: Level 2 administrative unit code, from GADM

GID_0
: Level 0 administrative unit code, from GADM. Corresponds to Iso3c country codes

GEZ_TERM
: Global ecological zone, from Geonetwork 2007. NA if data are not available.

un
: Unsdcode for the country, generated by the “countrycode” R package

Economic Characteristics

PCGDP.l

Lagged per capita GDP in constant year 2010 USD, from World Development Indicators. Corresponding code is NY.GDP.PCAP.KD. NA if data are not available.

PCGDP.change.l

Lagged change in per capita GDP in constant year 2010 USD, from World Development Indicators. Corresponding code is NY.GDP.PCAP.KD.ZG. NA if data are not available.

Pop.growth.l

Lagged change in population, from World Development Indicators. Corresponding code is SP.POP.GROW. NA if data are not available.

Ag.pct.l

Lagged agriculture, forestry, and fishing as a percent of national GDP, from World Development Indicators. Corresponding code is NV.AGR.TOTL.ZS. NA if data are not available.

Ag.emp.l

Lagged agricultural employment as a percent of total employment, from World Development Indicators. Corresponding code is SL.AGR.EMPL.ZS. NA if data are not available.

Political Characteristics

n_obs

For the nationally aggregated data, this is the number of forested cells in that country in that year. NA if data are not available.

polity2

Polity2 score of the country where the forested cell is located, from PolityIV dataset (Marshall et. al 2016). NA means a Polity2 score is not available.

Polity_class

Generated using the Polity IV dataset polity2 scores (Marshall et. al 2016). “Autocracy” if Polity2 < -5, “Anocracy” if Polity2 >= -5 and Polity2 <=5, “Democracy” if Polity2 > 5. NA means a Polity2 score is not available.

pr

1 if any component of the national electoral system uses a proportional representation system, 0 otherwise. Corresponds to the “pr” variable in Cruz et. al 2016. NA if data are not available.

democracy_BX

1 if the country is a democracy as defied by Boix et. al, 0 otherwise. Corresponds to “Democracy_BX” variable in Graham and Tucker 2016. NA if data are not available.

election_DPI

1 if an election year (either legislative or executive or both) , 0 otherwise. Cruz et. al 2016, calculated using “legelec” and exelec variables. NA if data are not available.

close80

1 if margin.norm>=80 (which corresponds to a margin of victory of <= 20 percent), NA if margin.norm<80 or data are not available, 0 if not an election year.

close90

1 if margin.norm>=90 (which corresponds to a margin of victory of <= 10 percent), NA if margin.norm<90 or data are not available, 0 if not an election year.

votediff

Difference between seat shares of the party which won the most seats and the party which won the second most seats. from Coppedge et. al 2020. Calculated as the difference between “v2ellostsl” and “v2ellostss”. NA if data are not available.

maj

Margin of victory (seatshare) for the government parties. Corresponds to the “MAJ” variable in Cruz et. al 2016. NA if there is no election or data are not available.

margin

Difference between percent of votes received by the government parties and the opposition parties. Calculated as the difference between the “numvote” and “oppvote” variables in Cruz et. al (2016). NA if there is no election or data are not available.

margin.norm

0-100 where 100 represents the most competitive election and 0 represents an election where one side received all of the seats/votes, as calculated using the votediff variable. NA if there is no election or data are not available.